

The Impact of Agricultural Lending on Women's Entrepreneurship in Kazakhstan's Agricultural Sector: The Role of Regional Factors and Access to Land

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Abstract

Women's entrepreneurship in agriculture is recognized as an important factor in inclusive growth; however, empirical data on the transition economies of Central Asia remain underresearched. This study analyzes the causal influence of institutional agricultural lending on the development of women-owned farms in 20 regions of Kazakhstan over the period 2015–2024 (175 observations). We employ a three-module analytical approach: (1) two-way fixed-effects panel regressions to identify causal relationships; (2) k-means clustering to account for regional heterogeneity; and (3) the random forest method to assess the predictive power of variables. Access to credit does not have a statistically significant causal effect on the number of female-headed farms in the region after controlling for fixed effects, despite a positive correlation in the pooled OLS model. The clustering results reveal two distinct regional groups: the first is characterized by a high level of female farming but relatively low growth, whereas the second starts from lower initial indicators but grows three times faster. In neither group do we find evidence that lending specifically drives this dynamic. Access to land emerges as the more significant factor: on average, women own only 2.01% of agricultural land (median 1.19%), which acts as an institutional constraint on female entrepreneurship. These findings indicate that simply increasing the volume of lending is insufficient and that differentiated, region-specific measures combining financial instruments with reforms in land rights and regional infrastructure are required.

Keywords: agricultural lending, women's entrepreneurship, financial inclusion, regional heterogeneity, Kazakhstan, inclusive development, land access

1 Introduction

Women's entrepreneurship in agriculture is increasingly recognized as a key driver of inclusive economic growth, rural development, and food security in both developing and transition economies (Quisumbing et al. 2023). Amid the transformation of agricultural systems, ensuring women's equal access to land, finance, and technology is becoming not only a matter of social justice but also a prerequisite for raising sectoral productivity (Meinzen-Dick et al. 2019; Diaz-Chavez 2025).

Kazakhstan retains a substantial state role in agricultural finance through institutions such as KazAgroFinance JSC, the Agrarian Credit Corporation JSC, and the Fund for Financial Support of Agriculture JSC. At the same time, the share of women heading farms remains low, and women's control over agricultural land is especially limited: on average, only 2.01% of agricultural land is registered to women, with a median of 1.19%.

The question of how access to finance affects farmers' performance has been widely studied in the literature (Tundui and Tundui 2024; Wasif et al. 2025). However, most empirical studies have at least three limitations. First, they rely on cross-sectional data and therefore do not clearly distinguish correlation from causation. Second, their geographical focus is concentrated in Africa and South Asia, while post-Soviet countries with distinct institutional legacies remain largely outside the evidence base. Third, there is a lack of research examining the relationship between lending and access to land. This combination may be key for women farmers, as our preliminary data show.

This study addresses these gaps by analyzing panel data for 20 regions of Kazakhstan over a 10-year period (2015–2024, 175 observations). The contribution of this work is threefold. First, it identifies the channels through which institutional agricultural lending may influence the development of women-owned farms in the Central Asian context. Second, it applies a three-module analytical design that combines fixed-effects econometrics, cluster analysis, and machine learning to disentangle interregional correlation, intraregional dynamics, and predictive importance. Third, it tests whether access to land acts as a binding institutional constraint that reduces the potential impact of financial support.

Based on the theoretical discussion and gaps in the literature, the study tests three hypotheses. H1: access to institutional agricultural lending has no statistically significant effect on the number of women-owned farms within regions. H2: the relationship between lending and women's entrepreneurial outcomes is heterogeneous and varies with regional infrastructure and institutional capacity. H3: access to land (land_share) positively moderates the effect of lending, but its institutional limitation reduces the overall effectiveness of financial support.

The empirical strategy is based on an unbalanced panel combining data from the National Statistics Bureau of the Republic of Kazakhstan, state agricultural financial institutions, and the Land Resources Management Committee. We estimate two-way fixed-effects models with regional clustering of standard errors, apply k-means clustering to identify stable regional regimes, and use random forest with group cross-validation to assess predictive power.

The remainder of the article is organized as follows. Section 2 reviews the literature and develops the conceptual framework. Section 3 describes the data, variables, and analytical framework. Section 4 presents the empirical results. Section 5 discusses the findings in relation to prior research. Section 6 develops policy recommendations for Kazakhstan's agricultural sector. Section 7 concludes.

2 Literature review: a critical synthesis

2.1 Theoretical foundations of inclusive development

Ensuring equal access to resources for all participants in agricultural production is defined in international studies as a strategy for inclusive development (Gartaula et al. 2025; Islam et al. 2024; Phillips et al. 2025). However, this general understanding masks significant disagreements. The question of exactly how inclusivity should be implemented is a subject of disagreement among authors.

Ros-Tonen et al. (2019) emphasize inclusivity in the value chain. Dabkienė (2025) notes that innovations in gender development are key methods for sustainable agricultural development. Doss (2025) highlights the crucial role of women's empowerment and its impact on household agricultural productivity.

One such point of contention concerns the role of partnerships between large agribusinesses and small-scale producers. Some researchers view these partnerships as the primary driver of inclusive growth (German et al. 2020; Schoneveld and Weng 2023). Others, on the contrary, show that in practice, large farms dictate the terms, putting small producers at a disadvantage (Schoneveld and Weng 2023).

An even more heated debate has emerged around technological innovations. Ye (2025); Nhelnyama (2026); Issahaku 2024; Han and Liang 2025; Yang and Li 2025) argue that digitalization and artificial intelligence can increase access to resources. Without accompanying institutional reforms, technological interventions are likely to exacerbate inequality, according to Emir et al. (2024), with the primary benefits going to the most resource-endowed producers. Our study takes a middle ground: we do not dismiss the potential of financial instruments, but we empirically test the conditions under which they are effective.

Issues related to public policy aimed at strengthening the digital financial services infrastructure in rural areas to ensure high-quality and sustainable growth in agricultural production are addressed in the study by Han and Zhang (2026). Modern approaches to agriculture require the integration of multidimensional models that take into account not only production indicators but also parameters of sustainability, inclusivity, and environmental integrity.

Many aspects of inclusive development are systematized in the work by Lidder et al. (2025), which identifies five key areas for the inclusive transformation of rural areas: (1) renewing long-term investment in research and development, taking into account the principles of equality and gender parity; (2) taking into account the interests of marginalized groups; (3) ensuring equal access to technologies; (4) leveling the conditions of competition by limiting the dominance of large corporations and supporting small and medium-sized agribusinesses; (5) prioritizing employment development in the context of automation and the expansion of value chains.

Kangasniemi et al. (2025), Mobarok et al. (2021), and Urago and Bozoğlu (2022) note that structural changes in rural regions depend on the inclusiveness of these processes. Climate impacts and low productivity are obstacles to the sustainable transformation of rural areas.

Two mechanisms that promote inclusive development – investment in human capital (healthy nutrition and education) and the equitable distribution of resources and reduction of risks for vulnerable populations – can lead to social well-being but require coordinated policy measures, according to [Chen et al. \(2026\)](#).

Its main areas of activity include: coordination of the agricultural and non-agricultural sectors, development of digital infrastructure in rural areas, and, which is equally important, the integration of principles of inclusivity into the modernization processes themselves.

An examination of the theoretical sources of inclusive agricultural development has allowed us to identify several key points.

There is no consensus on exactly how inclusivity should be implemented. Some authors emphasize value chains ([Ros-Tonen et al. 2019](#)), while others focus on innovation and the advancement of gender equality ([Dabkienė, 2025](#)) and on measuring women's empowerment and its impact on productivity ([Doss, 2025](#)). Using a three-module analytical approach, we emphasize the need for a comprehensive, multidimensional assessment of inclusivity.

The literature also reveals a contradiction in the assessment of the role of large agro-industrial enterprises. Partnership with large companies is often viewed as a driver of inclusive growth ([German et al. 2020](#)), whereas other studies show that large farms can dictate terms and exacerbate inequality among smallholder producers ([Schoneveld and Weng 2023](#)).

Discussion of technological and digital innovations follows a similar pattern. Some studies regard digitalization as a mechanism for reducing transaction costs and expanding access to resources ([Ye 2025](#); [Nhelnyama 2026](#); [Issahaku 2024](#); [Han and Liang 2025](#); [Yang and Li 2025](#)), while others argue that without institutional reform these tools may reinforce existing asymmetries ([Emir et al. 2024](#)).

[Lidder et al. \(2025\)](#) identifies five key areas for the inclusive transformation of rural areas, including addressing the interests of marginalized groups and ensuring equal access to technology.

A review of the theoretical framework revealed that inclusive agricultural development depends on institutional conditions, technological infrastructure, and social mechanisms. The identified contradictions justify the need for our study, which focuses on two key factors: access to credit as a financial mechanism and access to land as an institutional barrier.

2.2 Financial inclusion: correlation and causality

In existing studies, expanded access to financial resources is predominantly viewed as an important factor in entrepreneurial growth ([Yang and Li 2026](#)).

A positive correlation between the volume of lending and agricultural growth has been demonstrated in the following studies ([Tundui and Tundui 2024](#); [Wasif et al. 2025](#); [Namayengo et al. 2023](#); [Khan and Kim 2025](#); [Ullah et al. 2024](#); [Wang et al. 2025](#); [Mbangiswano et al. 2025](#)).

We identify three methodological limitations:

- Most cross-sectional studies are based on cross-sectional surveys, making it impossible to identify the causes of correlations;
- the authors of these studies and their geographical focus are concentrated primarily in African countries and South Asia ([Schoneveld and Weng 2023](#); [Mager and Faße 2023](#); [Ingutia and Sumelius 2024](#); [Nchanji et al. 2025](#); [Islam et al. 2024](#); [Wang et al. 2025](#); [Sassi](#)

2026). Post-Soviet countries, including Kazakhstan, with their unique institutional legacy, remain outside the scope of this analysis. However, their experience may differ significantly;

- Institutional constraints are overlooked. Research on financing rarely addresses access to land. However, for the agricultural sector, land is not only a productive resource but also a key collateral asset (Chebet 2023; Amoah 2024). Without taking this factor into account, the research is unlikely to be comprehensive.

However, there is one exception. Carter (2022) proposes flexible risk management tools tailored to households' resource capacities. Unfortunately, the question of whether this approach is applicable to women's entrepreneurship is not addressed.

Digital financial instruments deserve special attention. Researchers see potential in them: digital integration can reduce transaction costs and make financing more accessible. However, the extent to which this potential is realized for women farmers in post-Soviet countries has not been fully explored.

Mkhitarian (2021) and Omotayo and Omotoso (2025) identify several factors hindering the development of the agricultural sector. These include outdated technologies and farming methods, an inefficient financing system, unstable farm incomes, low liquidity of collateral, and a lack of technological and financial support. Each of these factors deserves separate attention.

The issue of where agriculture remains a priority is pressing for Kazakhstan. Government policy is focused not only on developing the financial sector; this year was declared the year of digitalization in Kazakhstan. This is being done to address issues such as the reallocation of resources in agriculture, infrastructure development, and expanding access to lending, primarily for small farms, through the use of digital inclusive finance as an alternative to traditional forms of support. These aspects are addressed in the works of many scholars (Lin and Peng 2025; Roberts 2026).

The key role of digital finance in promoting the sustainable development of rural regions was explored by Song et al. (2026). The authors demonstrate that investment in digital infrastructure is necessary to ensure the sustainable modernization of the agricultural sector.

In developing countries, modernization is often constrained by a combination of resource and institutional limitations. In response, comprehensive programs have emerged. Governmental and non-governmental organizations combine the provision of loans with training for farmers (OECD/GWEP 2025; Baffoe-Bonnie and Kostandini 2026; Nyberg et al. 2025).

These studies have a methodological shortcoming: they fail to distinguish between interregional correlation and intraregional causality. In this study, we propose a different methodological approach, using panel data for 20 regions of Kazakhstan over a 10-year period and a fixed-effects model (TWFE).

2.3 Gender barriers: an institutional approach

In our view, two competing approaches can be identified in the literature on gender development in agriculture.

The financial approach (Mazakaza and Odoyo 2022; Shahriar et al. 2025) emphasizes expanding access to microfinance and banking services as a key mechanism for women's empowerment. This finding demonstrates a positive correlation between access to finance and entrepreneurial outcomes.

Financial instruments are ineffective without addressing issues of land rights, market access, and participation in decision-making. This is a key feature of the institutional approach outlined in the works of (Quisumbing et al. 2023; Diaz-Chavez 2025; Adefila et al. 2024; Ge et al. 2023; Gil 2023; Mohammed et al. 2025). Credit alone cannot stimulate sustainable growth.

Barriers to access to land and credit were identified by (Diaz-Chavez 2025; Chebet 2023; Amoah 2024), who compared the relative importance of financial and institutional constraints and justified the need for land rights. However, the institutional environment in which they are studied differs from that existing in the Kazakhstan's economy.

In summary, ensuring gender equality and social integration in agriculture can be described as follows:

From production processes to social relations and institutional mechanisms is the subject of academic debate (Quisumbing et al. 2023; Timu et al. 2024; Agarwal et al. 2025). Successful inclusion requires taking the gender dimension into account throughout the entire process, from planning to monitoring.

We found a study of a systemic approach closely linked to inclusive agricultural development and financial support instruments for various agricultural groups, including women, in the article by Khalatur et al. (2023). Other authors, such as S (2022), characterized women's entrepreneurship as "family business". The promotion of women's active participation in leadership roles, leading to a reduction in social inequality in rural areas through the use of special measures, is reflected in the work of Rubin et al. (2019).

We identify two key points in these studies. The advancement of women, who possess a diverse range of skills and knowledge, contributes to economic stability, innovation, and the promotion of sustainable development in rural areas (Rubin et al. 2019). Ogunisola et al. (2024), and Adefila et al. (2024), supplemented the first point with the need for access to land, productive resources, finance, markets, and the creation of added value through cooperatives.

Gender equality not only drives innovative and social processes in agriculture but is also linked to the green economy and effective governance policies. This is highlighted by the following authors, including Lopez et al. (2022); Fertő and Bojnec (2024); Amoak and Najjar (2026); Zhou and Zhang (2026).

Farms managed by women are more likely to adopt agroecological practices and create sustainable food systems, thereby receiving more subsidies, increasing crop yields, and demonstrating a high level of autonomy, according to Mottet et al. (2020), and Kolinets and Tokar (2019), a view with which we cannot help but agree.

Stepanenko et al. (2023) add another important dimension. Sustainable development in agribusiness is impossible without equal access to clean natural resources, socioeconomic benefits, and social protection. In their view, this is a crucial aspect of inclusive rural development.

All of this leads to one conclusion: gender-oriented policies and support programs are not merely supplementary but a necessary condition for inclusive and sustainable agricultural development. Chebet (2023) empirically confirms this conclusion. The study shows that access to land rights has a positive impact on productivity and income but is often overlooked in policy-making.

Secured land rights empower women by enabling them to invest in agriculture and adopt improved farming practices. For Kazakhstan, where systemic barriers to women farmers persist, these findings have direct practical implications.

It should also be noted that women's access to microfinance ([Mazakaza and Odoyo 2022](#)) leads to their empowerment, improved living conditions, and better decision-making; therefore, it is necessary to conduct a more in-depth study of socio-economic issues and ways to address them.

According to [Stawicka and Parlinska \(2021\)](#), other factors motivating rural women to start their own businesses include a desire for independence, fair pay commensurate with the amount of work, and entrepreneurial skills.

In addition, women-led microenterprises, which form the backbone of rural communities and contribute to economic growth and sustainable development, often face significant challenges, particularly in securing financial support and establishing stable social networks ([Behera et al. 2026](#)).

According to [Mozumdar et al. \(2025\)](#) and [Jemaneh and Shibeshi \(2023\)](#), women's empowerment is linked to increased agricultural productivity and poverty reduction. Building on this idea, [Tundui and Tundui \(2024\)](#), [Shahriar et al. \(2025\)](#), and [Bohn et al. \(2024\)](#) note that access to lending and household economic status, as well as support for women entrepreneurs, have a positive and significant impact on entrepreneurial success.

[Ghosh and Neogi \(2018\)](#), [Costa et al. \(2026\)](#), and [Meinzen-Dick et al. \(2019\)](#) identify two key barriers for women entrepreneurs in agriculture: limited access to finance and inadequate infrastructure. Without addressing these, the authors conclude, business growth and sustainability are unlikely.

[Wasif et al. \(2025\)](#) demonstrate just how high the potential of agricultural entrepreneurship is. This includes income, food security, sustainable development, overcoming gender stereotypes, and job creation. The list is impressive.

But realizing this potential, as noted by [Demedeme and Opoku \(2022\)](#), and [Satpayeva et al. \(2020\)](#), is hindered by the same obstacles: limited access to financing, education, and training. Until these barriers are removed, the positive effects described by [Wasif et al. \(2025\)](#), will remain largely theoretical.

[Shali Lasrado et al. \(2025\)](#), and [Chance and Abdoul \(2025\)](#), emphasize that women's entrepreneurship is not merely a social issue. It is a key factor in economic development and the empowerment of women, particularly in rural areas.

In summary, it can be noted that the issue of gender inequality stems from fundamental systemic constraints, which can be categorized as follows: access to land ([Hartarska et al. 2024](#); [Amoah 2024](#); [Yan et al. 2022](#); [Lambrecht et al. 2023](#); [Baloi et al. 2023](#)) and women's participation in governance and decision-making ([Ragsdale et al. 2018](#); [Henrotte and Van den Broeck 2026](#); [Asadullah and Kambhampati 2021](#); [Ayamga et al. 2023](#); [Naheed and Rukhsana 2024](#)). The findings of this literature review confirm the validity of an institutional approach to analyzing women's entrepreneurship in the agricultural sector.

2.4 Regional heterogeneity: a gap in the literature

Most studies on financial inclusion assume that the impact of lending is universal. It either “works” or “does not work,” regardless of the region, infrastructure, or institutional context.

However, [Chen et al. \(2026\)](#) and [Cavatassi et al. \(2025\)](#) point out an important nuance. The effectiveness of financial instruments largely depends on additional investments in infrastructure, education, and market institutions.

The problem is that this aspect remains understudied. Two reasons can be identified. The first is methodological. Studies using cluster analysis to identify regional development patterns are extremely limited.

The second is empirical. Studies combining econometrics and machine learning methods applied to the agricultural economy are virtually nonexistent.

Our study aims to fill this gap. We integrate three analytical modules – TWFE, k-means, and random forest. This allows us not only to identify regional heterogeneity in Kazakhstan but also to assess the robustness of the results obtained.

Aspects of the issues related to the impact of credit on women farmers in the Kazakhstan economy, which has its own specific characteristics, remain understudied, which reinforces the relevance of our research. We examined methodological approaches to assessing the impact of various types of credit on the economic situation of women in agriculture based on the works of [Awuor et al. \(2021\)](#), [Agarwal \(2018\)](#), [Fenet Belayand and Alemayehu Oljira \(2019\)](#), [Huyer et al. \(2024\)](#), and [Mare \(2017\)](#).

There are also few studies by domestic researchers on this topic. The impact of women's entrepreneurship on economic growth was examined in the works of [Akybayeva et al. \(2024\)](#), [Bekturganova and Kurmasheva \(2025\)](#), and [Kuldasheva and Ahmad \(2025\)](#).

We integrated the technological and institutional approaches examined in the literature review to assess how lending interacts with an institutional factor such as women farmers' access to land. Our study differs from previous ones in that we attempt to empirically test whether credit support from financial institutions in Kazakhstan can compensate for limited access to land. Or does access to land remain a binding constraint that reduces the impact of lending?

3 Research methodology

The systematic logic of the study's design follows a three-stage sequence. First, the presence of a causal relationship is tested through panel econometric analysis ([Chiu et al. 2023](#); [Lee and Wooldridge 2023](#); [Wooldridge 2025](#)). Second, the structural context of the phenomenon is examined through cluster analysis. Third, the predictive significance of the results is assessed through machine learning methods.

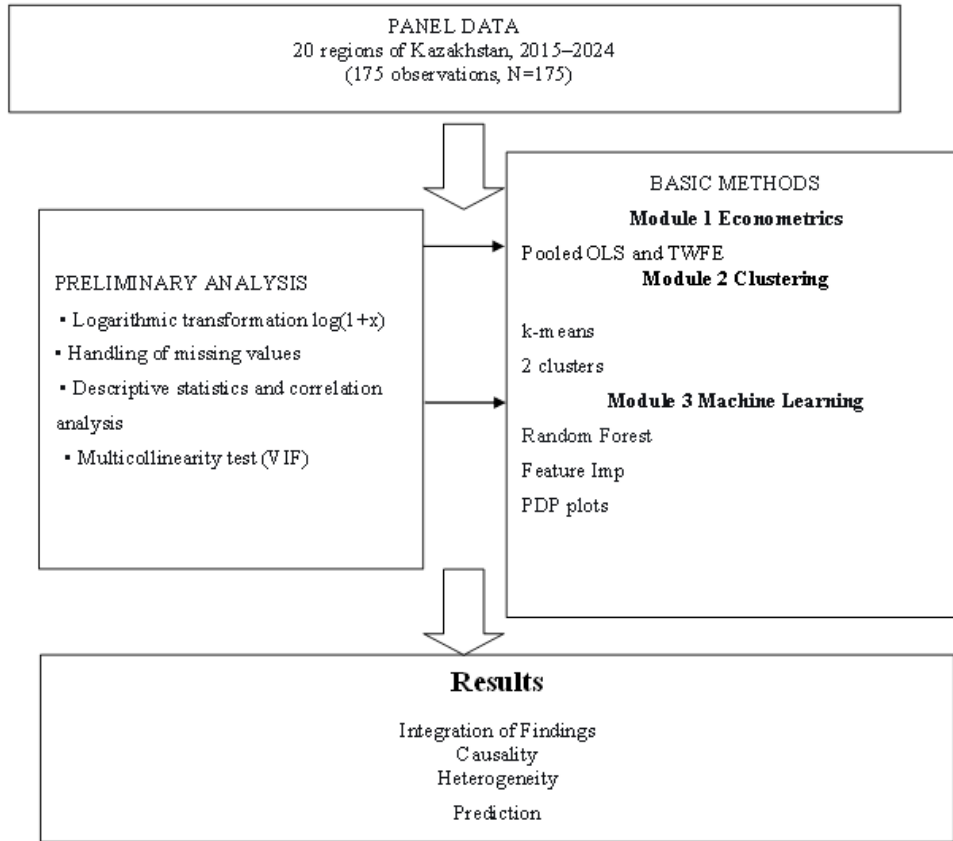


Fig. 1 Research design: three-module analytical framework

The analysis is based on unbalanced panel data for 20 regions of Kazakhstan covering the period 2015–2024, totaling 175 observations. The spatial unit is the region, and the temporal unit is the year.

Data sources: National Statistics Bureau of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan <https://stat.gov.kz/>:

- Data on women-owned farms and agricultural lending: KazAgroFinance JSC, Agrarian Credit Corporation JSC, and the Agricultural Financial Support Fund JSC for 2015–2024: (Women and men of Kazakhstan. Statistical compendium)
- Land use data from the Land Resources Management Committee of the Ministry of Agriculture of the Republic of Kazakhstan: “Proportion of women provided by agricultural land, broken down by form of ownership” (2015–2024).

The dataset includes variables characterizing the number of women-owned farms, the volume and sources of agricultural lending, as well as women’s access to land resources. A detailed description of the variables is presented in Table 1.

3.1 Description of variables and transformation

A logarithmic transformation was applied to prepare the data for econometric modeling. The transformation $\log(1 + x)$ was used for the variables `women_farms` and the lending indicators. This approach normalizes the distribution and allows the regression coefficients to be interpreted as elasticities (the percentage change in the dependent variable for a 1% change in the independent variable). The choice of the $\log(1+x)$ transformation is due to the presence of zero values in some components of lending.

The dependent variable is defined as:

`log_women_farms` – the logarithm of the number of active peasant farms headed by women.

Key independent variables:

`log_credit_total` – the logarithm of the total volume of institutional agricultural loans issued to women (the sum of the three components).

`log_credit_kaf` – the logarithm of loans issued by KazAgroFinance JSC.

`log_credit_acc` – the logarithm of loans issued by Agrarian Credit Corporation JSC.

`log_credit_fund` – the logarithm of loans issued by Agricultural Financial Support Fund JSC.

`lag1_log_credit_total` – first-order lag of the total lending volume variable (previous year's value) to assess the lagged effect.

Control variables:

`land_share` – the share of agricultural land allocated to women (as a percentage of the total agricultural land area in the region), which is important for distinguishing between the effects of credit and land ownership.

Identifiers:

`region_id` - numerical region identifier (1–20).

`year` - observation year (2015–2024). (Data is presented in Table 1)

Handling of missing values:

An analysis of data completeness revealed the presence of missing data at various levels. Missing data is acceptable when using a fixed-effects model in an unbalanced panel sample (Wooldridge 2023). The proportion of missing values for the `credit_fund` variable is high (39%) because no loans were issued between 2015 and 2018 in the cities (Astana, Almaty, Shymkent) and the newly established regions (Abai, Zhetysu, Ulytau). Due to the specific nature of the statistical registration of land use rights, the proportion of missing values for the `land_share` variable is 14.86%. The proportion of missing values for the `credit_total` variable is 2.86%.

3.2 Analytical modules

3.2.1 Module 1: Identifying causal relationships (fixed-effects panel regression)

To determine the impact of lending, we used panel econometric modeling, which allowed us to account for regional heterogeneity.

Basic model

In the first stage of our study, we estimate the relationship between variables without considering the data structure (Equation 1), using the combined pooled OLS model, which treats panel data as a single cross-sectional sample.

$$\log(\text{women_farms})_{it} = \beta_0 + \beta_1 \log(\text{credit_total})_{it} + \beta_2 \text{land_share}_{it} + \mu_{it}$$

(1)

Fixed-effects model

We also use a two-factor fixed-effects model (TWFE) to account for regional characteristics that remain constant over time (geography, infrastructure, and institutional conditions) and general macroeconomic conditions (Equation 2). This is a standard tool for identifying causal relationships in panel data.

To identify causal relationships in panel data, we applied the two-factor fixed-effects model (TWFE) (Equation 2)

$$Y_{it} = \alpha_i + \lambda_t + \beta X_{it} + \mu_{it} \quad (2)$$

Where:

Y_{it} – dependent variable (number of women-owned farms in region i in year t)

α_i – fixed effect of the region (regional-specific characteristics)

λ_t – fixed effect of the year (time-specific effects)

μ_{it} – random error term

Pooled OLS compares regions with one another. If the coefficient of variation is significant, this suggests that larger loan amounts are associated with increased output, i.e., there is a relationship. However, this model does not reveal the underlying causes.

TWFE compares each region with itself and shows the impact of lending on product output. With this in mind, we have selected this model.

A comparison of the pooled OLS and TWFE models shows that if a variable is significant in the first model but loses significance in the second, the observed correlation is explained by structural differences between regions rather than the level of lending support. In other words, large amounts of allocated credits are not the cause of growth in the regions' gross agricultural output. Other institutional factors are at play here, which we explore further.

To determine the impact of loans from different institutional sources and to test the robustness of the results, we divided the overall credit indicator into three components ($\log_credit_kaf$, $\log_credit_acc$, $\log_credit_fund$ – loans from KazAgroFinance, Agrarian Credit Corporation, and the Financial Support Fund) and performed separate regression calculations.

3.2.2 Module 2: Analysis of regional heterogeneity by clustering

We applied cluster analysis using the k-means algorithm. Regions were grouped based on seven characteristics:

- the mean value of $\log_women_farms$;
- standard deviation of $\log_women_farms$;
- mean value of $\log_credit_total$;
- standard deviation of $\log_credit_total$;
- average land share;
- linear trend of $\log_women_farms$ (slope coefficient);
- coefficient of variation of $\log_women_farms$.

All variables were standardized (z-score) prior to clustering. The optimal number of clusters ($k=2$) was determined by maximizing the average silhouette score, which was 0.400. Before clustering, we standardized all variables to ensure equal weighting (otherwise, loans in millions of tenge would have outweighed land share in percentages). We then determined the optimal number of clusters using the silhouette score. After examining options ranging from 2 to 5, we obtained the maximum value at $k = 2$. A silhouette coefficient of 0.400 indicates that dividing the data into two clusters makes sense: regions within each group are similar, but they differ between groups (Januzaj et al. 2023). The reliability of the solution was verified by 100 iterations of the algorithm with different initial centers. This approach allowed us to determine whether the relationship between lending and women's farming differs across regions with similar structural characteristics.

3.2.3 Module 3: Machine learning for predictive power and nonlinearity assessment

In addition to the causal inferences derived from econometric models, we applied the random forest algorithm (Salman et al. 2024). The objective was to assess the relative predictive power (importance of features) of lending variables and the land access variable, as well as to explore potential nonlinear relationships that may not be apparent in linear panel models.

The model was trained to predict `log_women_farms` using the following set of features: `log_credit_total`, `log_credit_kaf`, `log_credit_acc`, `log_credit_fund`, `lag1_log_credit_total`, `land_share`, `year`.

Hyperparameter optimization was performed using a random search method with 5-fold cross-validation. To do this, we optimized the following: the number of trees – 500 (`n_estimators`), the maximum depth limited to 5 (`max_depth`), the minimum number of samples for node splitting – 5 (`min_samples_split`), and the minimum number of samples in a leaf – 2 (`min_samples_leaf`). The split quality metric was the mean squared error (MSE), and the number of features considered at each split was set to the square root of their total number (`max_features`).

We applied Group K-Fold cross-validation with 5 folds to ensure a rigorous assessment of the model's ability to generalize to regions not included in training (the newly formed Abayskaya, Zhetysuskaya, and Ulytauskaya regions), where grouping was performed based on the region identifier `rtgion_id` (Mohapatra et al. 2020). This provides a conservative estimate of the model's predictive ability and ensures that data from the same region do not end up in both the training and test sets simultaneously. It also allowed us to verify how well the results can be generalized to the newly formed regions.

We evaluated the model's quality using the root mean square error (RMSE) and the coefficient of determination R^2 on the test folds. Two approaches were used to interpret the model. The first approach involved assessing the importance of features based on the average reduction in the root mean square error when a variable was excluded. The second approach involved partial dependency graphs for key variables (loans and land). These graphs show the average effect of changing a variable on the predicted value of the target variable while holding all others at their mean levels.

4 Synthesis of methods and model comparison strategy

All calculations were performed in the R environment (version 4.2.2) using the following packages: plm (panel econometrics), fixest (fixed-effects models), cluster (cluster analysis), randomForest (random forest), caret (cross-validation), and ggplot2 (visualization).

4.1 Analysis of panel data for 20 regions of Kazakhstan

An analysis of panel data for 20 regions of Kazakhstan covering the period 2015–2024 revealed significant regional variation (Table 1). The average number of women-owned farms is 3,127, but the range varies from 14 to 20,551, indicating extremely uneven development of women’s entrepreneurship in the agricultural sector.

Table 1 Descriptive statistics of key variables

oprule Indicator	Mean	Std. Dev.	Min	Median	Max	N
women_farms	3,127.45	4,298.91	14.00	1,440.00	20,551.00	175
credit_total (million KZT)	3,983.52	6,236.45	7.80	2,641.15	73,059.00	170
credit_kaf (million KZT)	905.43	1,134.92	4.00	428.00	5,597.00	155
credit_acc (million KZT)	2,726.15	6,281.00	0.00	1,201.00	72,963.00	153
credit_fund (million KZT)	1,119.20	630.52	129.30	941.00	2,811.00	107
land_share (%)	2.01	2.14	0.01	1.19	12.10	149
log_women_farms	7.34	1.24	2.71	7.27	9.93	175
log_credit_total	7.76	1.14	2.17	7.88	11.20	170

Given the high interregional variability (standard deviation of 6,236.5 million tenge), the average volume of institutional agricultural lending amounts to 3,983.5 million tenge. We observe that the highest percentage of missing values is found in the credit_fund variable (39%) (loans from the JSC “Financial Support Fund”). The reason is that in 2015–2018, these loans were not issued in cities of national importance (Astana, Almaty, Shymkent) or in new regions (Abai, Zhetysu, Ulytau).

At the same time, a critically important observation is the extremely low share of land allocated to women (an average of just 2.01%, with a median of 1.19%). This is a more serious barrier than access to loans, meaning that in most regions, women do not own land. Thus, although the number of farms and the volume of loans vary, the analysis shows that women’s limited access to land remains one of the main limiting factors.

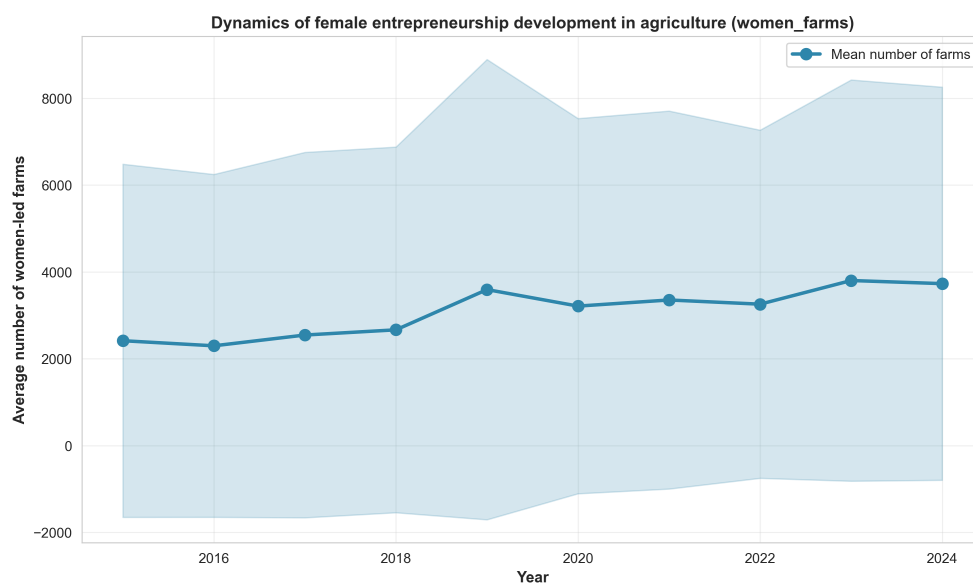


Fig. 2 Average number of women-owned farms, by region of Kazakhstan, 2015–2024

Although the volume of loans increased from 3 billion tenge to 4.5 billion tenge, this growth was concentrated in regions with more developed infrastructure (Figure 2). Econometric analysis confirms this conclusion. Pooled OLS summary data show a positive correlation between the number of farms owned by women and the volume of loans. However, when accounting for regional fixed effects using TWFE, this correlation disappears. The main conclusion is that financial institutions such as KazAgroFinance, the Agrarian Credit Corporation, and the Agricultural Financial Support Fund focus on supporting developed regions.

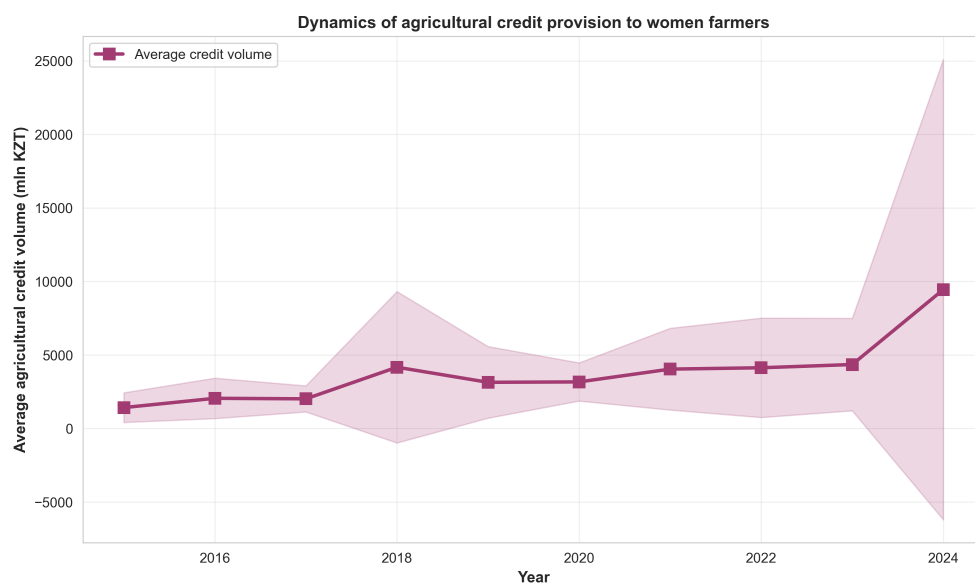


Fig. 3 Volume of loans issued to women farmers

The scatter plot (Figure 3) shows a significant correlation between the volume of institutional lending and the number of farms owned by women ($r = 0.576$, $p < 0.001$). Regions with high levels of lending are characterized by a large number of farms. Estimation of the two-way fixed-effects (TWFE) model showed that after accounting for regional differences, the coefficient of the lending variable decreased from 0.725 to 0.044 and lost statistical significance ($p = 0.576$). This implies that lending flows to developed regions but does not stimulate women's entrepreneurship within those regions. This raises the question: are loans effective, or are there other factors at play?

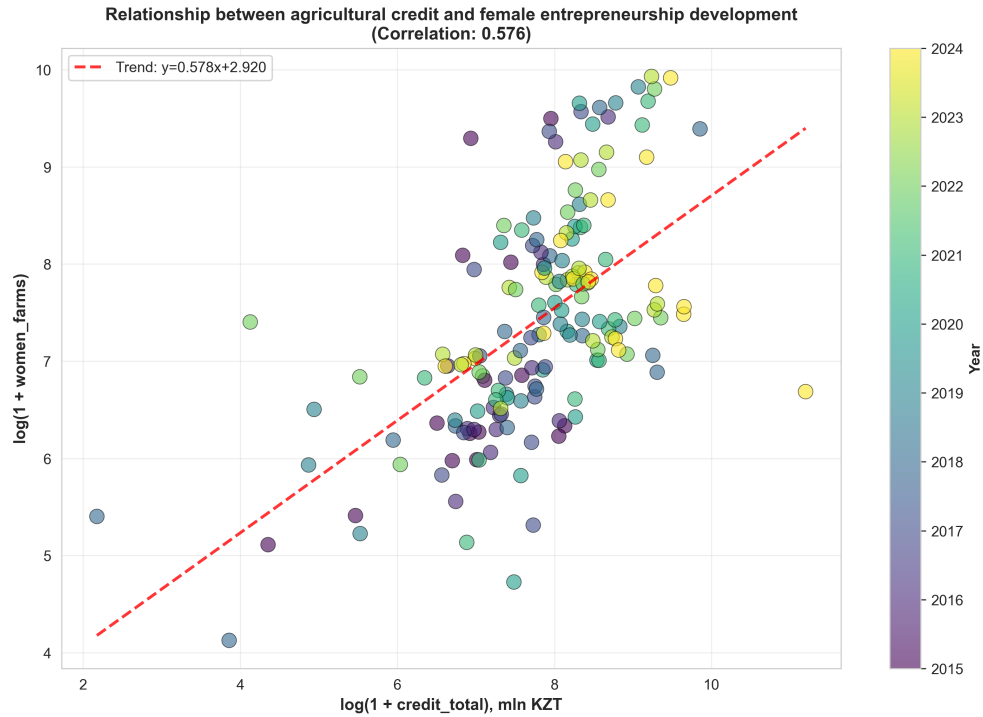


Fig. 4 The relationship between lending and women's entrepreneurship in agriculture

Figure 4 illustrates this relationship visually. Correlation analysis alone does not allow us to draw conclusions about the causal nature of this relationship. The observed relationship may reflect the influence of third factors, such as the region's level of economic development, the quality of the institutional environment, or the availability of infrastructure. In our view, the observed correlation can be explained by other regional factors or by a feedback mechanism, whereby the growth of women's entrepreneurship increases the demand for loans.

4.2 Panel regression results: econometric analysis

We constructed a panel regression model to assess the impact of institutional lending on the development of women-owned farms.

Stage 1: Ordinary least squares (Pooled OLS)

The pooled ordinary least squares (OLS) method revealed a positive correlation between total credit volume and the number of female-headed farms. The coefficient for the `log_credit_total` variable was 0.725 ($p < 0.001$). This is explained by the fact that a 1% increase in credit leads to a 0.73% increase in the number of female-headed farms. $R^2 = 0.336$, meaning the model explains 33.6% of the interregional variation in the dependent variable. The land access variable `land_share` is also significant at the 10% level, confirming the importance of land rights for women.

Table 2 Comparison of pooled OLS and TWFE

Variable	Coefficient	Standard Error	Coefficient	Standard Error
	Pooled OLS		TWFE	
log_credit_total	0.7250***	(0.0901)	0.0443	(0.0834)
land_share	0.0566*	(0.0341)	0.0181	(0.0289)
Constant	1.7143**	(0.6916)	-	-
N	149		149	
R^2	0.336		0.030 (within)	
Adjusted R^2	0.327		0.099 (overall)	
Fixed effects	No		Region + Year	
SE type	HC1-robust		Clustered (region)	

The pooled OLS model showed that the higher the volume of lending, the greater the number of farms. However, this is because these regions initially have better infrastructure and access to the loans. In other words, the specification of this model does not account for these regional characteristics.

Stage 2. TWFE Fixed-effects model

To account for these differences, we applied the TWFE fixed-effects model, which incorporates these regional characteristics. Here, we observe that the coefficient for the credit variable decreases from 0.725 to 0.044 and loses statistical significance ($p = 0.596$).

The following conclusions can be drawn: the second TWFE model primarily reflects interregional differences, meaning that loans are initially concentrated in regions with higher levels of women's entrepreneurship rather than stimulating its growth in regions. This finding further confirms the hypothesis that financial instruments are unable to stimulate the growth of women's entrepreneurship without corresponding institutional reforms, including expanding women's access to land.

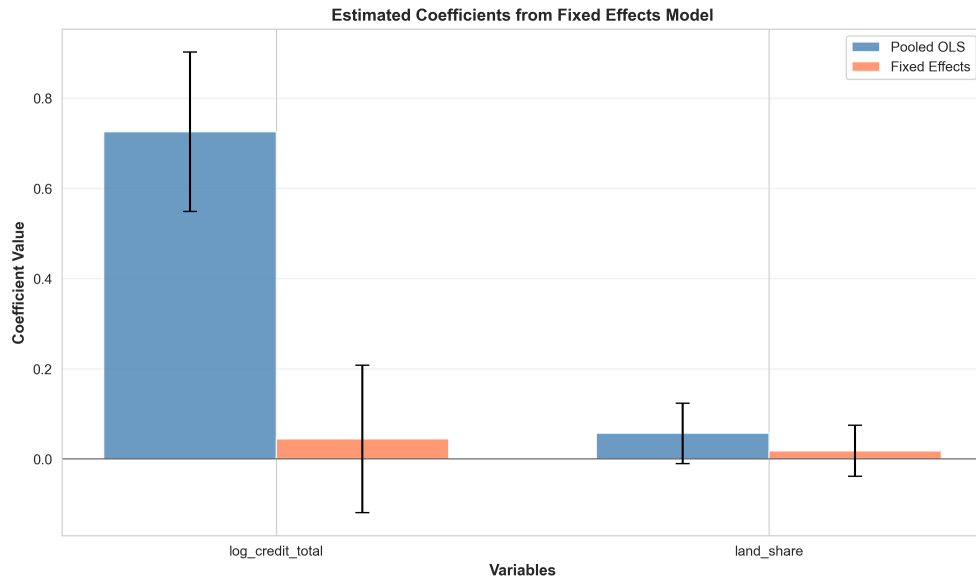


Fig. 5 Comparison of the pooled OLS and TWFE models

Figure 5 clearly shows that in the pooled OLS model, the credit coefficient is positive at 0.725 ($p < 0.001$) and significant. In the TWFE model, it is practically zero (0.0443) and crosses the zero line on the graph. This confirms that the observed interregional correlation is due to interregional differences, rather than the impact of lending on women's entrepreneurship.

Stability test by decomposing lending into sources

We replaced the aggregate lending indicator with three components: loans from KazAgro-Finance JSC, Agrarian Credit Corporation JSC, and the Agricultural Financial Support Fund JSC.

Table 3 Results of the model estimation by lending source

oprule Variable	Coefficient	Standard error	t-statistic	p-value	N
log_credit_kaf	0.0031	(0.0365)	0.084	0.933	88
log_credit_acc	-0.0473	(0.0484)	-0.979	0.331	88
log_credit_fund	-0.0073	(0.0647)	-0.113	0.910	88

The calculations presented in Table 3 show that none of the three sources of lending had a statistically significant effect, while the sample size was reduced to 88 observations due to missing data on lending from the Agricultural Financial Support Fund. The coefficients are close to zero, ranging from -0.473 to 0.003, and the p-values exceed 0.05 (0.933, 0.331, and 0.910). These calculations demonstrated once again that credit flows do not influence the formation of female-headed households (data from the main TWFE model)

4.3 Regional clustering

To identify regional heterogeneity in the characteristics of women's entrepreneurship development, the k-means clustering method was applied. Based on seven indicators – mean values and standard deviations of the number of women-owned farms and the volume of lending, the share of land, and the development trend – regions were grouped into two clusters (optimality determined by the Silhouette Score criterion = 0.400), as shown in Table 4.

Table 4 Regional cluster profiles

Cluster	Number of regions	log_women_farms (mean)	log_credit_total (mean)	Share of land (average, %)	Growth trend (annual)
Cluster 0	15	7.71	8.02	2.30	0.091
Cluster 1	5	6.35	6.75	0.18	0.296

Note: Average characteristics of regional clusters. Growth trend – the slope of the log_women_farms linear trend over time, characterizing the growth rate of women's entrepreneurship.

Cluster analysis divided the regions into two groups.

Cluster 0 (15 regions) – rural areas with a high level of women's farming (log_women_farms = 7.71, corresponding to approximately 2,230 farms on average per region), high total credit volume (log_credit_total = 8.02, or about 3,050 million tenge), and a relatively high share of land owned by women (2.30%). However, the growth rate here is moderate: 0.091 logarithms per year, which translates to about 9.5% annual growth.

Cluster 1 (5 regions) – regions with a low starting point. Here, there are four times fewer women-owned farms (log_women_farms = 6.35, about 570 farms), 3.6 times fewer loans (log_credit_total = 6.75, about 850 million tenge), and the share of land owned by women is practically zero (0.18%). However, the growth rate is three times higher: 0.296 logarithms per year, or about 34.5% growth per year.

Note: Since the dependent variable is taken in natural logarithm, the linear trend coefficient β corresponds to the approximate growth rate in percentage terms. For Cluster 0, $\beta = 0.091$ yields about 9.5% growth per year; for Cluster 1, $\beta = 0.296$ yields about 34.5% per year.

According to our calculations, these results are significant for two reasons. First, rapid growth in regions with low levels of lending suggests that loans are not the driving force. The dynamics of women's farming are not determined by the current volume of loans or the availability of land. Regions with the least initial opportunities demonstrate the greatest growth potential, while more developed regions are experiencing a saturation effect.

The results are consistent with the findings of fixed-effects models.

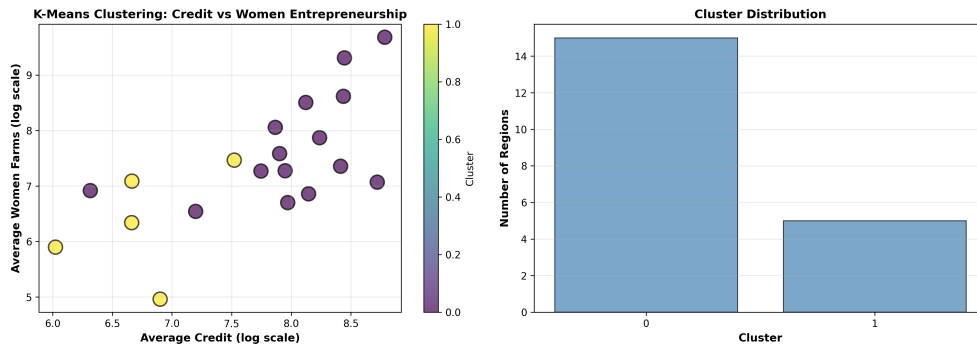


Fig. 6 Regional clusters based on lending and women's entrepreneurship

Figure 6 shows the regional cluster analysis results. In the scatter plot, Cluster 0 (blue) is located in the high-value area along both the lending axis and the women's entrepreneurship axis, while Cluster 1 (yellow) is in the lower-value area for both characteristics. Regions with a low baseline but high growth potential (cluster 1) require different measures than regions where the women's agribusiness sector is already developed (cluster 0). In our opinion, for the first group of regions in Kazakhstan, the priority may be the removal of institutional barriers (access to land, infrastructure), while for the second group, it may be improving efficiency and diversification.

4.4 Analysis of features' importance: random forest

To understand which variables best influence the development of women's farming and to check for nonlinear relationships, we used the random forest method. The model was trained using group cross-validation (GroupKFold, 5 folds per region). This ensures that data from the same region are not included in both the training and test sets.

Table 5 Feature Importance from the Random Forest Model

opruple Feature	Importance	% Total
log_credit_fund	0.448	44.76%
log_credit_total	0.146	14.56%
log_credit_acc	0.102	10.23%
lag1_log_credit_total	0.102	10.22%
land_share	0.088	8.84%
log_credit_kaf	0.082	8.19%
year	0.032	3.20%

We used a random forest model to identify the variables that best predict the development of women's agriculture and to test for nonlinear relationships that are not detected by linear models. We also used group cross-validation to assess the model's quality.

The data in Table 5 show that loans provided through the Agricultural Financial Support Fund account for the largest share at 44.8%. Total loans rank second at 14.6%, nearly three times smaller. This demonstrates the importance of the funding source and that not all loans are effective. It is crucial to distinguish loans by source; that is, the effectiveness of loans is determined by the source, not the volume. Systematic changes over the years do not affect the dependent variable. Land is also important for the sustainable development of agriculture and women's entrepreneurship (8.84%).

The random forest model showed that the variables can accurately predict outcomes and confirmed a significant correlation between loans and women farmers. The TWFE model, on the other hand, estimates the change in the outcome when each factor changes. This suggests that loans are concentrated in areas where women's entrepreneurship was already developed. This distinction between correlation and causation is the result of our study.

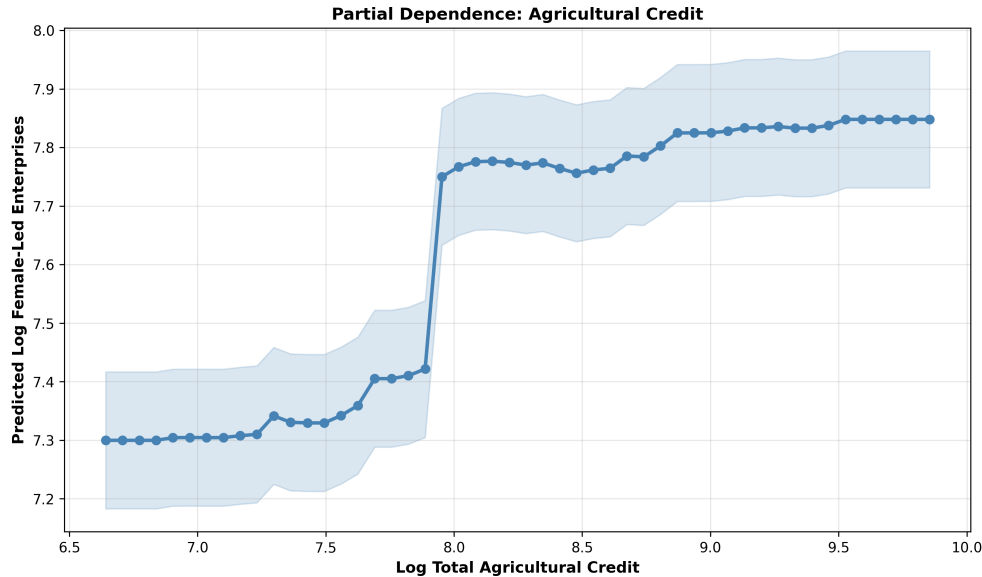


Fig. 7 Relationship between loan amount and the number of farms owned by women

The graph generated using the random forest model (Figure 7) shows a positive correlation between loan amount and the predicted number of farms owned by women. The absence of a significant effect in the TWFE model indicates that the relationship observed in Figure 7 is driven by the structural characteristics of the regions, rather than a causal effect of loans.

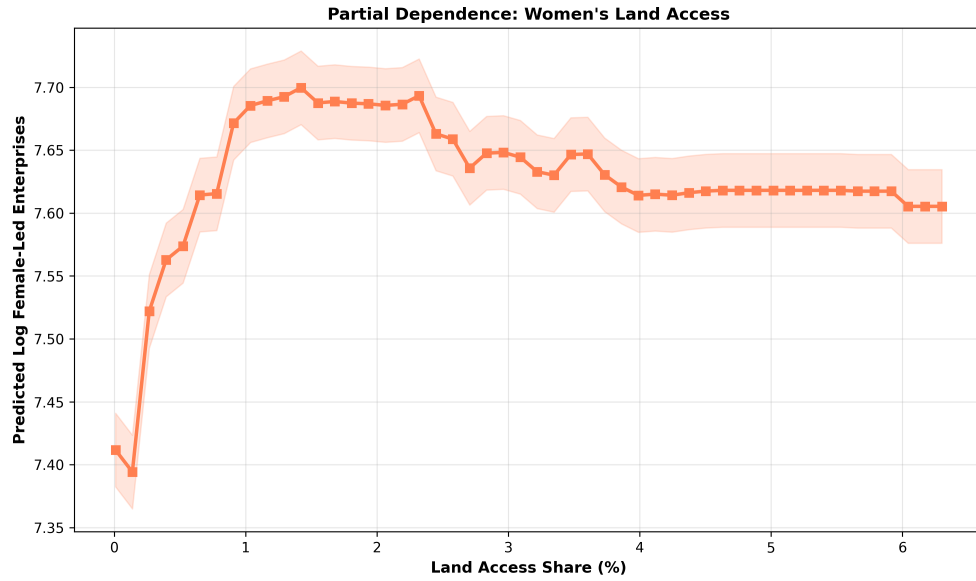


Fig. 8 Relationship between access to land and the number of women-owned farms

Figure 8 shows a nonlinear relationship between land ownership and the predicted number of farms owned by women, which is consistent with the hypothesis (Lin and Peng 2025; Han and Zhang 2026; Ye 2025). This relationship is weakly positive up to 2%, stabilizes in the range from 2% to 6%, and increases again above 6%. For Kazakhstan, this means that to enhance the impact of lending, it is necessary to overcome the 2% threshold in land ownership. Overcoming this barrier will lead to a multiplier effect and the use of land as a financial asset, consistent with the “critical minimum of assets” hypothesis for entrepreneurial activity described by Carter (2022).

However, the quality of the random forest model was low: $RMSE = 0.95$, $R^2 = -0.56$. The negative coefficient of determination indicates that the model predicts worse than the simple average. This suggests that the random forest model cannot adequately capture regional characteristics without introducing fixed effects. This confirms our main conclusion that lending is not the cause of the growth in the number of farms owned by women. Consequently, a more complex model, such as TWFE, is needed, one that accounts for regional heterogeneity and demonstrates that lending has no impact on the development of farms owned by women.

An analysis of panel data for 20 regions of Kazakhstan covering the years 2015–2024 reveals several key findings.

1. Lending does not have a negative impact on the growth in the number of farms owned by women. A simple pooled OLS model shows a positive relationship: a 1% increase in lending is associated with a 0.73% increase in the number of farms ($p < 0.001$). When applying time- and region-fixed effects (TWFE), the coefficient drops to 0.044, meaning it becomes insignificant ($p = 0.596$). Consequently, the correlation we observed is not explained by the

fact that loans support women's businesses in Kazakhstan, but rather by the fact that they are primarily directed toward more developed regions.

2. The result is robust. We disaggregated the loans by source (KazAgroFinance, Agrarian Credit Corporation, Financial Support Fund), but none yielded a significant effect. The sample under study was reduced to 88 observations (there is less data for the Financial Support Fund), but all three components yielded the same result. These results showed that lending does not play a significant role in the emergence of new women-owned farms.
3. Access to land is an institutional factor in the successful development of women's entrepreneurship in agriculture. The share of land held by women farmers averages just 2.01% (median 1.19%), and in some regions it is close to zero. This finding indicates that without resolving the issue of access to land, credit support alone cannot stimulate the growth of women's entrepreneurship in Kazakhstan's agricultural sector.
4. Two clusters of rural development emerge from the analysis. Cluster analysis (k-means) revealed a consistent division of regions into two groups. Cluster 0 (15 regions) is characterized by a high level of women's farming ($\log_women_farms = 7.71$) and lending ($\log_credit_total = 8.02$). It has relatively higher access to land for women farmers (2.3%), but a moderate growth rate of 9.5% per year. Cluster 1 (5 regions) exhibits low baseline levels ($\log_women_farms = 6.35$, $\log_credit_total = 6.75$), extremely limited access to land (0.18%), but growth of approximately 34.5% per year. Regions with low starting levels are characterized by rapid growth, but this growth is not linked to the current volume of loans, which also remains low in these regions (confirming the absence of a uniform lending effect in TWFE).
5. Loans effectively support but do not influence the development of agriculture in Kazakhstan. Analysis using the random forest method showed that variables related to loans explain 87.9% of the model's predictive power. Loans from the Financial Support Fund are particularly significant (44.8%). The partial dependency graphs confirm that the more loans there are, the higher the predicted number of female-headed farms. However, the model generalizes poorly to the newly formed regions of Kazakhstan ($R^2 = -0.56$). In other words, loans effectively explain in which regions there are more farms (interregional differences), but they do not explain why they are growing (intra-regional dynamics). Our calculations showed that loans are concentrated in regions where women's agriculture was initially well-developed.
6. The findings have direct implications for Kazakhstan's agricultural policy and demonstrate the ineffectiveness of universal approaches in supporting women's entrepreneurship. A broad expansion of lending without taking into account structural regional differences and institutional barriers (especially access to land) will not lead to an increase in women's entrepreneurship in agriculture.

5 Discussion of the results

5.1 Interpretation of key findings: lending constraints

The main finding of the study is a comparison of the estimates obtained using the pooled OLS and TWFE models. While the first specification shows a statistically significant positive effect of lending, the second demonstrates that this effect disappears entirely after controlling for fixed effects of region and year. This observation is consistent with the institutional approach to sector development (Quisumbing et al. 2023; Schoneveld and Weng 2023). Loans, provided

mainly in areas with a high share of women entrepreneurs, do not contribute to the creation of new farms. This is confirmed by data obtained using two additional analytical methods.

1. Cluster analysis identified two types of regions: developed regions with high levels of female entrepreneurship and access to loans (15 regions) and less developed regions (5 regions) with high growth rates. In both groups, the level of lending does not determine growth trends.
2. Random forest analysis confirmed that variables characterizing lending are indicators of regional differences; however, the negative correlation coefficient (R^2) from cross-validation showed that unobserved regional characteristics predominate. Taken together, the data confirm hypothesis H1: the observed correlations reflect the structural heterogeneity of regions (between variables) rather than a causal effect of lending on intraregional dynamics (within variables).

5.2 Research results. So, why don't loans work?

The main finding of the study that we wish to highlight is the discrepancy between the two models: pooled OLS (with a credit coefficient of 0.725) and the TWFE model (0.044). The first shows a positive effect of loans, while the second shows their complete disappearance after accounting for differences between regions (geography, climate, infrastructure) and time-fixed effects (price increase, changes in government development programs, macroeconomic trends, and others). This observation, consistent with the institutional approach to inclusive development (Quisumbing et al. 2023; Schoneveld and Weng 2023), suggests that the lack of an effect in the region is due not to the ineffectiveness of lending programs, but to structural factors.

Carter (2022) and Lin and Peng (2025) have shown that digital financial instruments can reduce transaction costs but are ineffective without access to human capital and collateral. In Kazakhstan, the share of women who own land is 2.01%, which is a critical threshold.

Issahaku (2024) reaches a similar conclusion in his study of digital finance and agriculture: access to financing is a necessary but insufficient condition. Without accompanying institutional reforms (land rights, market access, advisory services), lending measures will not lead to sustainable growth.

Lending is widespread in regions with a developed female entrepreneurship sector, but it does not contribute to the creation of new business forms. This is confirmed by data obtained using two additional analytical methods:

1. Cluster analysis identified two stable regional types in Kazakhstan: developed regions with high levels of women's entrepreneurship and access to loans (15 regions) and less developed regions with low indicators but high growth rates (5 regions) (cluster 1).
2. Lending variables are the most important factors effectively reflecting differences between regions (random forest algorithm). However, the negative coefficient of determination (R^2) from the cross-validation indicates that unobserved regional characteristics contribute significantly to the variation in the dependent variable of credit.

The results confirm Hypothesis H1: the observed relationship between lending and the number of farms owned by women reflects structural heterogeneity across regions (interregional differences) rather than a causal effect of lending on intraregional dynamics (intraregional differences). Lending is concentrated in regions where farms owned by women are initially more developed, but it does not stimulate their growth in lagging regions.

The random forest algorithm identifies differences between regions, while the TWFE algorithm shows that credit is ineffective within regions. Hypothesis H1 is confirmed: there is a correlation, but no causal relationship.

5.3 Land as a factor in institutional constraints on the sustainable development of women's entrepreneurship in Kazakhstan

In Kazakhstan, the share of agricultural land owned by women farmers remains critically low (mean 2.01%, median 1.19%). This finding correlates with Hypothesis H3 and confirms the position of [Chebet \(2023\)](#) and [Amoah \(2024\)](#): in the agricultural sector, land functions as a productive asset and as collateral. Lending support cannot fully realize its potential without adequate guarantees ([Diaz-Chavez 2025](#); [Adefila et al. 2024](#)). Institutional barriers neutralize the potential impact of financial instruments, which requires a reevaluation of agricultural policy priorities.

5.4 Regional characteristics of support policies for women farmers in Kazakhstan

The results of the cluster analysis confirmed Hypothesis H2: the impact of credit resources is not universal but depends on regional conditions. Regions with a low initial level (cluster 1) demonstrate accelerated growth, but this growth does not depend on the volume of available loans, indicating the effectiveness of financial support programs in Kazakhstan's regions.

These results are consistent with the findings of [Chen et al. \(2026\)](#) and [Li and Zhang \(2025\)](#) regarding the need for infrastructure investment. Based on the clusters, we recommend the following:

For cluster 0 (developed regions): Rather than increasing the volume of loans, priority should be given to improving the efficiency of existing resource utilization.

For cluster 1 (lagging regions): A comprehensive approach is needed, combining financial support with institutional reforms (land rights, infrastructure, etc.).

Compared to studies in South Asia ([Islam et al. 2024](#)) and Africa ([Doss 2025](#)), the problem of "limited access" due to land-based collateral is widespread, but in Kazakhstan it is further exacerbated by the specific nature of land relations. Unlike in African countries, where microcredit supplements financial resources, in Kazakhstan, lending still requires land as collateral.

5.5 Reliability of results: methodological aspects

The reliability of the results is ensured by the integration of three analytical modules, each of which makes a unique contribution to the overall study of the development of women's entrepreneurship in Kazakhstan's agriculture:

TWFE module. The main result does not reveal the cause of the impact of loans within regions. This calculation demonstrates that credit support without corresponding institutional reforms does not yield the expected results.

K-means module. Cluster analysis identified two stable regional systems, demonstrating that regions differ from one another (in terms of access to resources, level of development, institutional conditions, infrastructure, etc.).

Random forest module. Credit-related variables proved to be important for explaining differences between regions, demonstrating that key development factors are linked to regional characteristics which remain constant over time, rather than to lending support.

Based on this, it can be concluded that access to lending is a necessary but not sufficient condition for the expansion of women's entrepreneurship in Kazakhstan's agricultural sector.

5.6 Study limitations and directions for future research

In analyzing the results, we identified several limitations that may have affected the accuracy of our conclusions:

Large sample size. We examined regional data rather than data on individual districts or farms managed by women. This may have led to the omission of some important local characteristics (Gil 2023).

Missing data. Information on loans is missing (approximately 39% of the data). This is because in some years, certain (newly established) regions were not supplied with loans (Financial Support Fund).

Land registration. We recorded only the percentage of allocated land, without specifying whether it belonged to the enterprise or was simply used by it (Lambrecht et al. 2023).

Causal relationship. Our model shows a correlation between loans and business development, but it does not guarantee that loans stimulate growth. The opposite may be true: successful enterprises are more likely to apply for loans. More sophisticated methods are needed to obtain a more accurate answer, which requires a separate study.

6 Implications of the results for improving agricultural policy in Kazakhstan

The key finding – that a broad-based increase in agricultural lending does not stimulate the growth of women's entrepreneurship without addressing institutional constraints – calls for a reevaluation of existing approaches to financial support. Specific recommendations, tailored to different levels of government and types of regions, are outlined below.

6.1 Recommendations at the national level

Revision of lending programs. Current institutional lending programs (KazAgroFinance JSC, Agrarian Credit Corporation JSC, Agricultural Financial Support Fund JSC) are primarily focused on volume-based indicators of loan disbursement. The results of TWFE models demonstrate that this approach does not have a causal impact on the development of women's entrepreneurship. It is recommended:

To introduce gender-sensitive criteria for evaluating applications: simplified application procedures, flexible repayment schedules, and consideration of the seasonal nature of agricultural work;

To develop alternative collateral mechanisms: cooperative guarantees, crop insurance, and the use of movable property as collateral;

To monitor the distribution of loans by recipient gender and publish annual reports to ensure transparency.

The integration of financial support with land reform. Since access to land is a key institutional constraint (on average, 2.01% of land is held by women), lending interventions must be accompanied by measures to expand women's land rights:

To simplify the procedures for registering land rights for women, including reduced state fees and advisory support;

To encourage the transfer of land shares to women as part of privatization and state land redistribution programs;

To develop land lease mechanisms that prioritize women farmers, including subsidies for lease payments.

Inter-agency coordination. The effectiveness of financial policy depends on the coordination of actions between the Ministry of Agriculture, the Ministry of Finance, and regional akimats. It is recommended to establish an inter-agency working group on gender inclusion in the agricultural sector with a mandate to develop comprehensive support programs.

6.2 Differentiated recommendations by region type

The results of the cluster analysis identified two stable patterns of regional development, which justifies the need for a territory-oriented approach.

For regions in Cluster 0 (developed: 15 regions)

Characteristics: high baseline levels of women's entrepreneurship and access to credit, moderate growth rate (0.091 per year), relatively higher access to land (2.30%).

Policy priorities:

Improving resource efficiency: financial literacy programs, advisory support for business management, and encouraging cooperation to reduce transaction costs;

Diversification of activities: supporting the transition from traditional crop farming to processing, organic farming, and agritourism – that is, sectors with higher added value;

Digitalization: introduction of mobile platforms for accessing market information, online lending, and distance learning.

For Cluster 1 regions (catching up: 5 regions)

Characteristics: low baseline indicators for women's entrepreneurship and lending, extremely limited access to land (0.18%), but a high growth rate (0.296 per year).

Policy priorities:

Comprehensive support packages: a combination of start-up capital (microcredits, grants) with institutional reforms (access to land, infrastructure);

Development of basic infrastructure: investments in roads, logistics, storage facilities, and product collection points;

Pilot projects on land rights: launching experimental programs to simplify the registration of land rights for women, followed by scaling up successful practices;

Mentoring and knowledge sharing: partnership programs with regions in Cluster 0 to transfer knowledge and best practices.

7 Conclusion

This study presents a comprehensive analysis of the impact of institutional agricultural lending on the development of women-owned farms in 20 regions of Kazakhstan over the period 2015–2024. The application of a three-module analytical approach (panel econometrics, clustering,

machine learning) allowed us to draw conclusions about the nature of the relationship between lending and women's entrepreneurship in the agricultural sector.

7.1 Key findings

The observed positive correlation between the volume of lending and the number of women-owned farms ($r = 0.576$, $p < 0.001$) is driven by structural differences between regions, rather than by the causal impact of lending within regions.

After including regional (geography, infrastructure, climate) and time-varying fixed effects (price inflation, macroeconomic policy) (TWFE), the coefficient of the credit variable decreased from 0.725 to 0.044 ($p=0.596$), indicating the absence of a statistically significant causal effect. None of the three components of institutional lending (KazAgroFinance JSC, Agrarian Credit Corporation JSC, and Agricultural Financial Support Fund JSC) had a significant impact in the TWFE models, indicating the robustness of the main conclusion.

Access to land is a key institutional constraint. The average share of agricultural land owned by women is only 2.01% (median 1.19%). The land share variable was not significant in any of the TWFE specifications, indicating its role as a long-term link that neutralizes the potential impact of financial support.

Cluster analysis identified two stable patterns of regional development:

Group 0 (15 regions) – high levels of women's entrepreneurship and lending, moderate growth rates (0.091 per year);

Group 1 (5 regions) – low control indicators, but three times higher growth rates (0.296 per year), not linked to actual lending volumes.

Random forest machine learning methods confirm that credit variables dominate the structure of feature importance (87.9% overall). However, a negative R^2 (-0.56) in cross-validation between regions indicates limited generalizability of the model to new regions, which confirms the conclusion regarding the dominant role of regional differences.

7.2 Contribution to the research

This study contributes to the literature on financial inclusion and gender equality in agriculture in three ways:

Geographical novelty: For the first time, causal estimates of the impact of agricultural lending on women's entrepreneurship are presented in the context of Central Asia's transition economies, filling a gap in the literature focused on African and South Asian countries.

Methodological novelty: The combination of three analytical modules (TWFE, k-means, and random forest) allowed us to distinguish correlations between regions from intra-regional differences and assess the impact of lending on the development of women's entrepreneurship in Kazakhstan's agriculture.

Theoretical rationale: Empirically confirms the hypothesis regarding the interdependent nature of institutional constraints: land tenure neutralizes the potential impact of financial support, thereby supporting an institutional approach to inclusive development ([Quisumbing et al. 2023](#); [Schoneveld and Weng 2023](#)).

7.3 Directions for future research

Analyzing data from specific farms will allow for the consideration of regional heterogeneity and the individual characteristics of women's entrepreneurship.

Conducting interviews (surveys) and thematic studies can help identify institutional barriers (land rights, social norms) to access to lending.

7.4 Directions for improving Kazakhstan's agricultural policy

The study's findings have direct implications for agricultural policy in Kazakhstan and other transition economies:

- For Cluster 0 regions (developed): The priority is to improve the efficiency of existing resources through financial literacy programs, advisory support, and the promotion of cooperation.
- For Cluster 1 regions (catching up): Comprehensive programs are needed that combine financial support with institutional reforms, expanding women's access to land, developing infrastructure, and removing administrative barriers.
- At the national level: Lending programs need to be revised to account for gender-specific factors (simplified collateral requirements, flexible repayment terms) and the distribution of loans by recipient gender needs to be monitored.

This study demonstrates that financial inclusion alone cannot stimulate sustainable growth in women's entrepreneurship in the agricultural sector without accompanying institutional reforms. Lending is an important policy tool, but its effectiveness is determined by institutional conditions, primarily access to land and the quality of regional infrastructure. A differentiated, place-based approach that combines financial instruments with institutional reforms represents the most promising path toward inclusive agricultural development in Kazakhstan and Central Asia.

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Declarations

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Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

This research does not involve human participants and does not include human data, individual persons' data, or human tissue.

Consent for publication

Not applicable.

Data availability

The data used in this study was obtained from publicly available official sources:

- National Statistics Bureau of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan: <https://stat.gov.kz/>
- Data on women-owned farms and agricultural lending (KazAgroFinance JSC, Agrarian Credit Corporation JSC, and Agricultural Financial Support Fund JSC) (Women and Men of Kazakhstan. Statistical Compendium)
- Data on land use from the Land Resources Management Committee of the Ministry of Agriculture of the Republic of Kazakhstan: “Proportion of women who have been granted agricultural land, broken down by form of ownership” (2015–2024). National Statistics Bureau of the Agency for Strategic Planning and Reforms of the Republic of Kazakhstan: <https://stat.gov.kz/ru/sustainable-development-goals/goal/5/>

All data is presented in aggregated form at the regional level and have been processed in accordance with the terms of use of the relevant sources.

Materials availability

Not applicable.

Code availability

Not applicable.

Author contribution

Aimurzina B. developed the concept, supervised the entire study, collected and analyzed the data, wrote the initial draft, and reviewed it. Arzayeva T. participated in the concept development, validation, writing, and editing. Bulakbay Zh. participated in the concept development, writing, and editing. Mataibaeva G. participated in the concept development, writing, and editing. Sadvakassov R. participated in the concept development, data processing, validation, and editing.

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